ProClosure: Hierarchical Room-Object Assignment using Progressive Boundary Closure from Monocular Video - Supplementary Material

A Reconstruction and Structural Segmentation

Reconstruction. Geometry comes from a feed-forward pointmap SLAM front end [21], which for each keyframe $I_t \in \mathcal{K}$ returns a dense per-pixel point map in a shared world frame, a per-pixel confidence, and the pose $T_t \in SE(3)$. Low-confidence pixels are discarded; the confident points of all keyframes are pooled, voxel-downsampled, and outlier-cleaned to give $\mathcal{P}$, which is monocular and hence fixed only up to a global similarity.

Segmentation and lifting. Each keyframe is segmented by a promptable open-vocabulary model [20] with a fixed prompt set, returning scored instance masks; low-scoring masks are rejected. A mask is lifted to 3D without a separate depth step, since the front end already gives a 3D point per pixel: the mask selects the points of its own keyframe, after removing non-finite, degenerate, or low-confidence points. Masks retaining too few valid points are dropped as too marginally observed to localise.

Structural channel. Masks for the structural prompts (*wall*, *window*) are not treated as objects; their points accumulate across keyframes into the structural cloud $\mathcal{P}_w \subset \mathcal{P}$, the boundary from which the top-down map is formed. $\mathcal{P}_w$ is recovered without depth supervision, and its coverage is uneven (Sec. 3.1).

Door-frame channel. Masks for the *door frame* prompt accumulate into a third cloud, kept apart from $\mathcal{P}_w$. A door frame is not boundary during the closure; it is boundary only where visibility is measured, so it is carried separately and projected onto the grid in Sec. 3.1.

Object channel. Masks for the remaining prompts yield object observations, each a voxel set, associated online into persistent tracks $O = \{o_j\}$. A new observation with voxel set A is matched, via a spatial index, only against tracks sharing its voxels, using the overlap coefficient

$$\mathrm{IoS}(A, B) = \frac{|A \cap B|}{\min(|A|, |B|)}. \tag{4}$$

IoS replaces IoU so that a small fragment largely contained in a track accumulated over many views still scores high rather than being penalised by the track's size; an observation matching two tracks bridges them. Each finished track reduces to a centroid c_j and an oriented box. These components are used as released; the stage is not a contribution and supplies only $\mathcal{P}_w$, the door frames, the trajectory T, and the object tracks with their centroids and oriented boxes.

B The Workflow in One Diagram

Figs. 2a and 2b of the main paper split the pipeline in two so that each half fits a column. Fig. 4 draws the same workflow in one piece, with every intermediate product and the closure sequence in a single view.

C Properties of the Closure

The room count is emergent: it starts as the number of components of Φ and is set by geometry, not supplied; the corrections of Sec. 3.2.3 may then discard a region or found one. *Each room is resolved at its own scale*: a small room behind narrow apertures and a large one behind a wide opening both close, at different iterations, which one radius cannot achieve. *Apertures are not classified*: the closure treats a doorway and an unobserved gap identically, needing only that an opening be narrow, which is the property that suits it to monocular reconstruction. Detected door frames enter only where visibility is measured, never as a partition criterion.

D Rooms and Cleanup

Regions that never sealed. A pocket can stop growing because it ran out of admissible cells, or because it ran into the edge of the raster. Only the first is a room; the second is capture lying outside the building, bounded by where the reconstruction stops rather than by a wall. The two are told apart by the building footprint Ω_{fp} of Sec. 3.1, the filled outer contour of the projected full cloud, which marks the area the reconstruction covers. A region is discarded when

$$\frac{\left|\mathcal{R}_i \setminus \Omega_{\mathrm{fp}}\right|}{\left|\mathcal{R}_i\right|} > \phi, \tag{5}$$

with ϕ a fixed fraction. Its seeds are kept and passed to the rule below.

Seeds that cannot see their region. A seed relocating under Eq. (9) can end up in a neighbouring space, where it witnesses the wrong region. We score each seed against the centroid c_i of every region with the agreement measure $a(\cdot, \cdot)$ of Sec. F.0.1. Seeds are visited in ascending order of $a(p, c_{i(p)})$, their agreement with their own region, so the worst-placed seed is resolved first; seeds released by the rule above have no region of their own and are visited before any of them. A seed whose own agreement reaches τ stays. Otherwise it moves to the region of highest agreement if that exceeds τ; if none does, the seed founds a region of its own. A region founded this way is immediately a candidate for the seeds still to be visited, so a seed belonging nowhere can gather the neighbours that belong with it.

Doorways as boundary. The third correction is stated in Sec. 3.2.3: every visibility test is computed on $W \cup \Delta$ rather than on W, so a fan cannot cross an open doorway. A seed whose own position falls on Δ sees nothing in any direction and is removed rather than re-homed, since it stands in the aperture and belongs to neither adjoining region.

Displaced seeds. A seed may slip through an aperture into an adjacent space before it seals, ending in the wrong region. Within

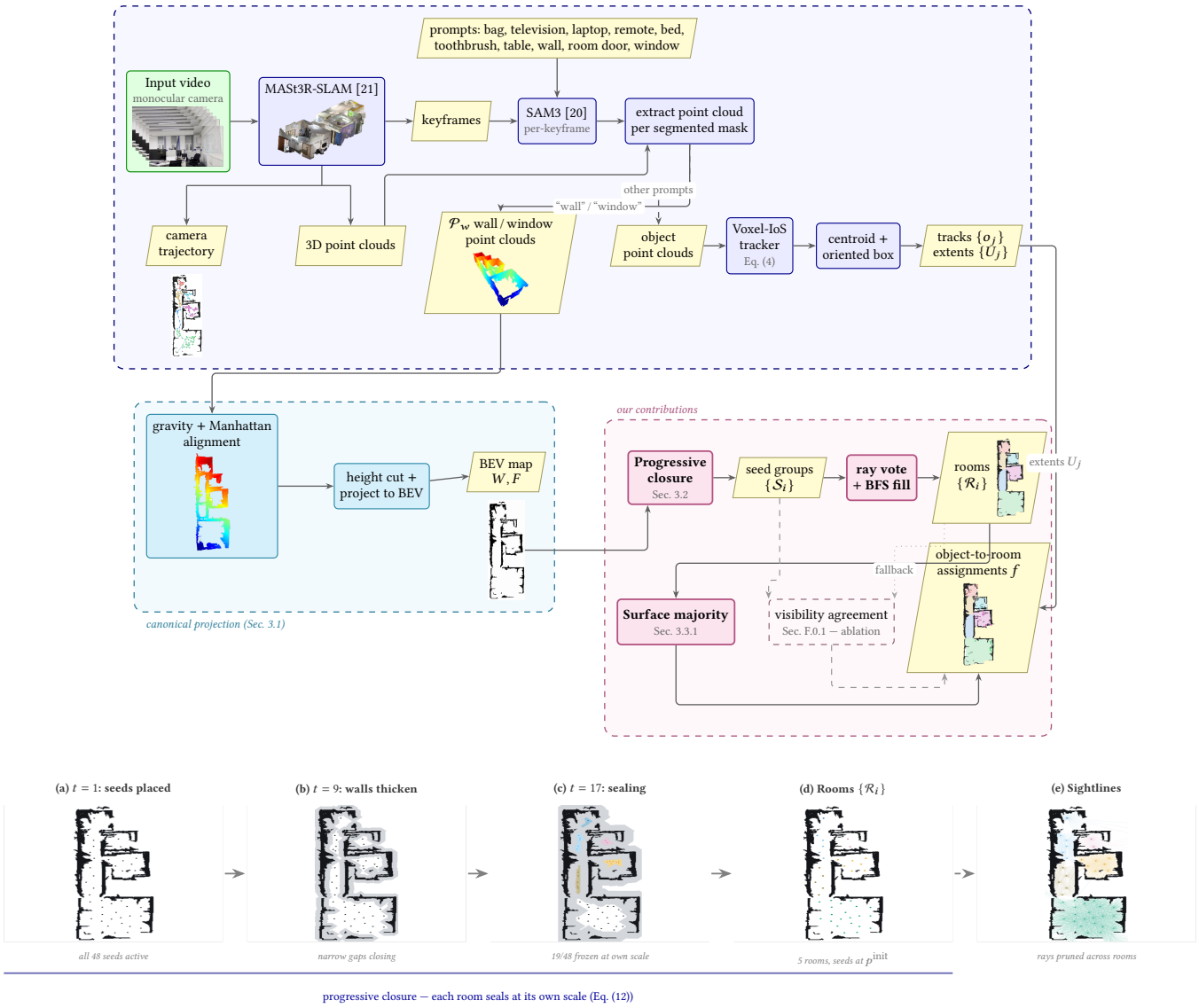


Figure 4: The full workflow in one diagram. A monocular RGB video is processed by MASt3R-SLAM [21], which returns keyframes, a dense point cloud and the camera trajectory. SAM3 [20] segments each keyframe against a fixed prompt set; masks for *wall* and *window* accumulate into the structural cloud $\mathcal{P}_w$, and the remaining prompts become object observations tracked by voxel-IoS matching. The structural cloud is aligned, height-cut and rasterised into a binary top-down map (W, F). Our contributions are shown in pink. *Progressive closure* grows the boundaries of W until every pocket saturates, seeded by the projected camera trajectory rather than by sampling the map (Sec. 3.2); its output is a set of seed groups $\{\mathcal{S}_i\}$, one per recovered room. The retained sightlines of those groups then vote for a label at each free cell, and a breadth-first fill restricted to free space completes the labelling, giving the dense rooms $\{\mathcal{R}_i\}$. The two assignment rules consume different products of this chain, which is why the flow branches at $\{\mathcal{S}_i\}$. *Surface majority* needs the dense labels: it assigns each track to the room holding most of its reconstructed extent U_j (Sec. 3.3.1). The dashed box is the *visibility-agreement* rule it is compared against in Sec. 5.3, which scores a track by the sightlines it shares with each room's seeds (Sec. F.0.1) and therefore requires no dense labelling at all; it consults the labels only through the dotted fallback edge, taken when no room reaches a nonzero agreement.

each room of at least $n_{\min}$ seeds we discard those whose displacement $\|p^{\text{init}} - p^{\text{final}}\|$ exceeds $\bar{d} + k\, s_d$ (mean and s.d. over the room's seeds); smaller rooms are left untouched.

Room representation. Freezing fixes how many rooms there are and which seeds witness each, at every room's own scale; it does not by itself fix their extent, since the frozen pocket is the region as of its seal and is eroded by the dilation that sealed it. A separate delineation pass (Sec. 3.2.4) recovers full extents, and those extents are what the room evaluation of Sec. 4 scores. Every seed is scored and casts its sightlines from its sampled position p^{init}, not from wherever relocation left it.

E Room Detection Algorithm

Algorithm 1 states the full Progressive Boundary Closure method for room detection.

F Object Placement Algorithm

Algorithm 2 states the visibility-agreement procedure of Sec. 3.3 in full.

F.0.1 Visibility agreement From a cell p we cast K rays at angles $\theta_k = 2\pi k/K$, each stopping at the first cell of $W \cup \Delta$, at the edge of Ω, or at a length cap ℓ, set separately for seed and object fans. Writing $r_k^o(p)$ for the kth ray of the object fan at p and $r_l^s(q)$ for the lth of the seed fan at q, agreement is the fraction of the object's rays that meet some ray of the seed's,

$$a(p,q) = \frac{1}{K}\sum_{k=1}^{K} \mathbb{1}\big[\, \exists\, l : r_k^o(p) \cap r_l^s(q) \neq \emptyset \,\big] \in [0,1]. \tag{6}$$

Two points inside one enclosed region see the same boundary, so their fans cross often and a is high. Two points separated by an intact wall see disjoint stretches of boundary and never meet. Across a narrow opening only the few rays passing through it can cross, so a stays small whatever the opening's cause, which is the property the rule relies on.

Each object is scored per room by mean agreement with that room's seeds,

$$s_i(o_j) = \frac{1}{|\mathcal{S}_i|}\sum_{q\in\mathcal{S}_i} a\big(\pi(o_j), q\big), \tag{7}$$

and is assigned to the highest-scoring room when that score reaches a threshold τ_a, falling back to the room of the nearest free cell otherwise. Appendix F gives the remaining details, and Sec. 5.3 reports the fallback rate and the comparison against surface majority.

G Evaluation Protocol in Full

The first is the *overlap* criterion established for this task [4] and adopted throughout the scene-graph literature [14, 28]. Writing $\mathcal{R}_e$ for the predicted rooms and $\mathcal{R}_g$ for the ground-truth regions,

$$P_{\text{ov}} = \frac{1}{|\mathcal{R}_e|}\sum_{r_e\in\mathcal{R}_e} \max_{r_g\in\mathcal{R}_g} \frac{|r_e \cap r_g|}{|r_e|}, \qquad R_{\text{ov}} = \frac{1}{|\mathcal{R}_g|}\sum_{r_g\in\mathcal{R}_g} \max_{r_e\in\mathcal{R}_e} \frac{|r_e \cap r_g|}{|r_g|}. \tag{8}$$

The maximum is taken independently in each direction, so the pairing is neither one-to-one nor thresholded. Low P_{ov} indicates under-segmentation and low R_{ov} over-segmentation.

Algorithm 1 Room detection by progressive boundary closure

Require: occupied set W; door frames Δ; footprint Ω_{fp}; trajectory T; parameters $\rho, \kappa, \sigma, \mu, t_{\max}, k, n_{\min}, \phi, \tau$
Ensure: rooms $\{\mathcal{R}_i\}$ with seed clusters $\{\mathcal{S}_i\}$

$\mathcal{S} \leftarrow$ TrajectorySeeds(T, W, σ, κ) ▷ Sec. 3.2.2
$\Phi \leftarrow \emptyset$; $W^{(0)} \leftarrow W$
for $t = 1$ **to** $t_{\max}$ **do**
  $W^{(t)} \leftarrow$ dilate $W^{(t-1)}$ by B_ρ, protecting Φ ▷ Eq. (10)
  $D^{(t)} \leftarrow$ EDT of $\Omega \setminus W^{(t)}$ ▷ Eq. (2)
  for all undeclared seeds p with $D^{(t)}(p) < \kappa$ (to a fixed point) **do**
    $p \leftarrow \arg\min_{q\in A^{(t)}(p)} d_g(p,q)$ if $A^{(t)}(p) \neq \emptyset$ ▷ Eq. (9)
  end for
  $V^{(t)} \leftarrow \{\, u \in F^{(t)} \setminus \Phi : D^{(t)}(u) \geq \kappa,\ d_{\mathcal{S}}^{(t)}(u) \geq \sigma \,\}$ ▷ Eq. (11)
  for all components C of $F^{(t)} \setminus \Phi$ with $C \cap \mathcal{S}^{(t)} \neq \emptyset$ **do**
    if $C \cap V^{(t)} = \emptyset$ **then**
      $\Phi \leftarrow \Phi \cup C$ ▷ saturated; Eq. (12)
    end if
  end for
  if all seeds lie in Φ **or** $W^{(t)} = W^{(t-1)}$ **then break**
end for
assign each seed to the component of Φ containing it, giving clusters $\{\mathcal{S}_i\}$
score every seed from its sampled position p^{init}
corrections (Sec. 3.2.3)
for all clusters i **do**
  if $|\mathcal{R}_i \setminus \Omega_{\text{fp}}|/|\mathcal{R}_i| > \phi$ **then**
    discard region i; return its seeds unassigned ▷ Eq. (5)
  end if
end for
$c_i \leftarrow$ centroid of region i; evaluate $a(\cdot,\cdot)$ on $W \cup \Delta$
for all seeds p in ascending order of $a(p, c_{i(p)})$, unassigned seeds first **do** ▷ $i(p)$: index of p's current region, so $c_{i(p)}$ is its centroid
  if p lies on Δ **then** discard p; **continue**
  if $a(p, c_{i(p)}) \geq \tau$ **then**
    continue ▷ sees its own region
  end if
  $i^\star \leftarrow \arg\max_i a(p, c_i)$
  if $a(p, c_{i^\star}) \geq \tau$ **then** move p to cluster $i^\star$ **else** start a new cluster at p
end for
for all clusters with $\geq n_{\min}$ seeds **do**
  discard seeds with displacement $> \bar{d} + k\, s_d$ ▷ outlier removal
end for
$\{\mathcal{R}_i\} \leftarrow$ Delineate($\{\mathcal{S}_i\}, W \cup \Delta$) ▷ Sec. 3.2.4
return $\{\mathcal{R}_i\}, \{\mathcal{S}_i\}$

Algorithm 2 Object placement by visibility agreement

Require: tracks O with centroids $\{c_j\}$; seed clusters $\{\mathcal{S}_i\}$; occupied set W; door frames Δ; grid Ω; projection π_Q; parameters $K, \ell_s, \ell_o, \tau_a$
Ensure: assignment f, route flag g

for all $q \in \bigcup_i \mathcal{S}_i$ **do**
  $R^s(q) \leftarrow$ cast K rays from q on $W \cup \Delta$; stop at $W \cup \Delta$, $\partial\Omega$, or ℓ_s steps
end for
for all $o_j \in O$ **do**
  $p \leftarrow \pi_Q(c_j)$
  $R^o(p) \leftarrow$ cast K rays from p on $W \cup \Delta$; stop at $W \cup \Delta$, $\partial\Omega$, or ℓ_o steps
  for all rooms i **do**
    $s_i \leftarrow \frac{1}{|\mathcal{S}_i|} \sum_{q \in \mathcal{S}_i} a(p, q)$ ▷ Eqs. (6), (7)
  end for
  $i^\star \leftarrow \arg\max_i s_i$
  if $s_{i^\star} \geq \tau_a$ **then**
    $f(o_j) \leftarrow i^\star$; $g(o_j) \leftarrow$ VIS
  else
    $q^\star \leftarrow \arg\min_{q \in \bigcup_i \mathcal{S}_i} \|p - q\|_1$
    $f(o_j) \leftarrow$ room of $q^\star$; $g(o_j) \leftarrow$ FALLBACK
  end if
end for
return f, g

The second is an *instance* criterion: predictions and regions are matched one-to-one by Hungarian assignment on IoU, a pair counts as a true positive when its IoU exceeds $\tau \in \{0.25, 0.5\}$, and we report $P_\tau = \mathrm{TP}/|\mathcal{R}_e|$, $R_\tau = \mathrm{TP}/|\mathcal{R}_g|$, $F_{1,\tau}$, and mean IoU over matched pairs. It is the stricter of the two, as it penalises a prediction that overlaps the right region loosely, which Eq. (8) rewards.

Scoring domain. Both criteria are computed on the cells of the top-down grid, the plane in which the partition is defined, whereas prior work computes them over 3D voxels or surface points. The difference is not cosmetic. Sampling the annotated mesh weights a room by the surface it contains rather than the floor it covers, so a cluttered bedroom carries several times the evidence of an empty hall of equal size; and 16.9% of the sampled surface carries no region identifier at all, mostly ceilings and upper walls. Scoring on the grid removes both effects. Surface-point figures appear alongside in Table 3 for comparability, and we state the domain wherever our numbers sit beside published ones. Scores are averaged over the ten floors, so each floor weighs equally and the means describe the same quantity as the tests below.

H Three Findings Across the Results

Three findings carry across Secs. 5.1–5.6. The first is that the room-count gap drives most of Table 2: our reimplementation of HOV-SG predicts 44 rooms for 72 observed regions where we predict 74, so its conservatism buys precision at the 0.25 threshold (0.98) at the cost of recall (0.60), while ours trades a little high-threshold precision for a partition close to the true count and a better F_1 at both thresholds. The second is that the doorway seal is the one stage whose contribution is separable at this sample size, improving all ten floors for +0.034 mIoU; it is also the one stage that consumes an input the baseline does not receive, which Sec. 4.3 states. The third is that the failures of Sec. 5.6 are structural rather than incidental, and the three floors discussed there are specific evidence for them rather than a generic caveat.

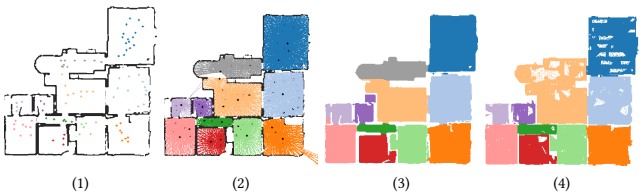


Figure 5: Phase A/B walkthrough, `00824-Dd4bFSTQ8gi` floor 0, final-method (camera-pose) seeding. 1: 83 seeds (of 85 recorded poses on this floor; the other 2 do not survive de-duplication), coloured by eventual cluster. 2: kept sightlines after cross-cluster pruning (Sec. 3.3). 3–4: delineated rooms vs. matched GT regions, same colouring. Nine of 11 predicted rooms match a GT region at both thresholds (P 0.82, R 0.90, F_1 0.86); mean IoU over matched pairs is 0.695, with one unmatched room.

I Phase A/B Walkthrough

Figure 5 follows one floor through the two phases of Sec. 3.2, from the seeds the trajectory supplies to the delineated rooms scored against the annotated regions.

J Room-Partitioning Families

Clearance-based methods locate rooms where the distance to the nearest obstacle is large. HOV-SG [28] thresholds this field, takes the surviving components as room cores, and floods outward from them; Hydra [14] maintains a graph of locations lying at safe distance from obstacles and partitions that graph into rooms; Point2Graph [31] detects regions from a density map with enhanced borders. All depend on clearance contracting at every passage between one room and the next.

Connectivity-based methods instead reduce free space to a skeleton or adjacency graph and sever it where it narrows [12, 25]; ROSE2 [17] first recovers dominant wall directions so that clutter does not produce spurious constrictions, and OccuSG [9] decomposes occupancy directly at narrow passages. Learned methods take a different route, regressing room polygons from a map or point cloud [7, 8, 16, 30, 32], at the cost of annotated training data and, in most cases, layouts close to rectilinear.

K Canonical Frame in Full

Gravity alignment. Planar surfaces are extracted from $\mathcal{P}$ by sequential RANSAC, each plane removed before the next is fitted. Among candidates whose normal is approximately vertical, the one with most inliers is taken as the floor, and $R_g \in SO(3)$ maps its normal onto the vertical axis.

Manhattan correction. A rotation about the vertical remains. Points in a central height band with near-horizontal normals are

treated as wall samples, their normals projected to the horizontal plane, reduced to orientations $\theta = \operatorname{atan2}(n_y, n_x)$, folded modulo $\pi/2$ and histogrammed. The modal orientation gives a yaw R_y that aligns the dominant wall direction with the grid, so $Q = R_y R_g$. This reduces aliasing of wall lines only; the closure of Sec. 3.2 is isotropic and requires no Manhattan layout.

Height selection. A boundary matters only where an agent can meet it, and the reconstruction is sparsest near the ceiling. We cut the transformed cloud at the mid-height of its own vertical extent and rasterise the half below, which spans the height envelope such an agent occupies. Expressed as a fraction of the scene's own height, the cut requires no metric scale.

L Further Limitations

Junctions with no constriction. The second mode is not a wide opening but the absence of any opening to find. On `00829-QaLdnwvtxbs` floor 0 the walkway and the room adjoining it share a frontier of 262 free cells, against 4 cells for a doorway elsewhere on the same floor and a median of 39 over all adjacent pairs. The two spaces meet along an entire wall line, with no door, no frame and no wall stub, so the growth has no width minimum to seal and the pocket spanning both never divides. The walkway seeds relocate into the room as the boundary thickens and the room is recovered as part of the walkway, at IoU 0.24 against 0.48 to 0.85 for every other region on that floor. Unlike the first mode this is not a matter of relative extent: the room is not shallow, and the architecture simply provides no minimum for a width-based rule to act on. Recovering such a boundary requires evidence the top-down map does not carry, such as a change in ceiling height or floor material.

Boundaries placed short. `00877-4ok3usBNeis` floor 1 is one of three floors on which HOV-SG scores higher (0.755 against 0.689). The room count is not the problem: we recover 8 rooms for 9 observed regions where HOV-SG recovers 3. Several of our boundaries fall short of the annotated ones, so rooms that are correctly separated overlap their regions loosely, and mIoU rewards a smaller number of well-fitted rooms over a larger number of approximate ones. It is also the floor on which visibility agreement most clearly beats surface majority (ARI 0.641 against 0.513, Sec. 5.3).

M Seed Relocation in Full

Relocation. Thickening the boundary encroaches on seeds. A seed whose region is not yet declared and whose clearance has fallen below κ moves to the nearest cell that is still admissible,

$$p' = \arg \min_{q \in A^{(t)}(p)} d_g(p, q),$$
$$A^{(t)}(p) = \left\{ \begin{array}{c} D^{(t)}(q) \geq \kappa, \\ q : \|q - p_u\| \geq \sigma \ \ \forall u \neq p, \\ d_g(p, q) \leq \mu \end{array} \right\}, \tag{9}$$

where d_g is geodesic distance within $F^{(t)} \setminus \Phi^{(t)}$ and μ caps the displacement. If no admissible cell is reachable the seed stays where it is. Two details matter. Distance is measured along free space rather than in straight line, so a seed cannot step across a wall to reach open space on the far side. Declared regions are excluded from the search, so a seed cannot drift into a room that has already been resolved.

N Progressive Closure in Full

Each step dilates the occupied set by B_ρ outside the declared cells and leaves the declared cells untouched:

$$\begin{aligned} W^{(t)} &= \big(\delta_{B_\rho}\big(W^{(t-1)}\big) \setminus \Phi^{(t-1)}\big) \cup \big(W^{(t-1)} \cap \Phi^{(t-1)}\big), \\ F^{(t)} &= \Omega \setminus W^{(t)}. \end{aligned} \tag{10}$$

A cell qualifies when it is at least κ from the boundary and at least σ from every current seed:

$$V^{(t)} = \big\{ u \in F^{(t)} \setminus \Phi^{(t-1)} : D^{(t)}(u) \geq \kappa,\ d_{\mathcal{S}}^{(t)}(u) \geq \sigma \big\}, \tag{11}$$

with $d_{\mathcal{S}}^{(t)}(u) = \min_{p \in \mathcal{S}^{(t)}} \|u - p\|_2$. A pocket is saturated when it holds no qualifying cell; every saturated pocket containing a seed is declared:

$$\Phi^{(t)} = \Phi^{(t-1)} \cup \bigcup_{C \in \mathcal{C}_{\text{new}}^{(t)}} C, \tag{12}$$

where $\mathcal{C}_{\text{new}}^{(t)} = \{ C \in \mathcal{C}^{(t)} : C \cap \mathcal{S}^{(t)} \neq \emptyset,\ C \cap V^{(t)} = \emptyset \}$ and $\mathcal{C}^{(t)}$ are the components of $F^{(t)} \setminus \Phi^{(t-1)}$.

O Ground-Truth Provenance

Ground truth requires no annotation on our part. Room extents come from projecting HM3D-Semantics region labels onto the same grid as our top-down map, and object-to-room membership is read from instance annotations that already carry their containing region. Room scores are computed over the regions the trajectory actually observed, since a region with negligible observed extent cannot be recovered by any method; this leaves **72** of the 105. Object scores use all 105, because an object's assignment is defined whether or not its region was well covered.

P Recurring Failure Modes

(1) *Wide openings.* A space joined to its neighbour by an opening comparable to its own extent does not enclose before the neighbour does, and the two are recovered as one. (2) *Junctions with no constriction.* Where two spaces meet flush, the growth has no width minimum to seal and they never divide. (3) *Unobserved regions.* Where the capture leaves a region largely unseen, the boundary is too sparse to enclose and no room is declared. Of the 105 annotated regions across the ten floors, 33 fall below the observation threshold of Sec. 4.2 and are excluded from the room evaluation. (4) *Alignment failure.* Where the recovered vertical is wrong, the top-down map is not a floor plan and the partition is meaningless. (5) *Track fragmentation.* A single physical object recovered as several tracks is assigned several times, which inflates the object count and can split one object across rooms.

Q Ablations in Full

Seed placement. Seeding from the camera trajectory gives mIoU 0.802 against 0.781 for farthest-point sampling, and is ahead on nine of the ten floors, the single exception losing by 0.064. The margin does not reach significance ($p = 0.064$) and we report it as a trend. Its source is the mechanism described in Sec. 3.2.2, in that a trajectory records where the agent stood and therefore places seeds

inside the rooms the walk entered rather than across an eligible pool. The procedural advantage is clearer: camera-pose seeding requires neither an eligibility pool nor a random selection step, since a recorded pose is free space by construction, and no random choice is made, so the segmentation is deterministic and admits no variance over seeds.

Doorway seal. Treating detected door frames as boundary wherever visibility is measured raises mIoU from 0.768 to 0.802 and improves all ten floors, giving $p = 0.002$, the smallest value the test can return for ten paired samples. The recovered room count moves from 71 to 74 against 72 observed regions. This is the only ablation reported here that separates its variants, and also the only one that relies on an input the baseline does not receive (Sec. 4.3).

Object-assignment rule. Surface majority gives ARI 0.696 against 0.689 for visibility agreement, at an identical room count since only the assignment stage differs. It leads on seven of ten floors, but $p = 0.193$, and on `00877` floor 1 visibility agreement is better by 0.128, which outweighs several smaller differences in the other direction. We adopt surface majority because it leads on most floors, because it requires no visibility threshold, and because it is considerably cheaper, the ray rule costing $n_{\text{obj}} n_{\text{seed}} n_{\text{ray}}^2$ segment tests per floor. We do not adopt it on the grounds that the two rules are separable at this sample size.

R The Three Corrections

Closure alone leaves three characteristic errors, and we correct each. A region bounded by the edge of the raster rather than by a wall never satisfies the enclosure criterion, and is discarded. A seed that cannot see the region it was assigned to is re-homed to the region it can see. Detected door frames are treated as boundary wherever visibility is measured, which prevents sightlines from crossing an open doorway; these are an additional input, supplied by SAM3 [20] in the monocular pipeline and by the dataset annotations on HM3D, and Sec. 5.3 reports what they contribute.

S Implementation Details

Reconstruction and segmentation. MASt3R-SLAM runs uncalibrated, single-threaded. SAM3 is queried per keyframe with a fixed 11 label prompt set: *wall* and *window* are the structural prompts of Sec. A ($\mathcal{P}_w$); the remaining eight (*bag*, *television*, *door frame*, *remote*, *table*, *bed*, *laptop*, *toothbrush*, *wall*, *window*, *room door*) are tracked as objects. Detections below score 0.8 are rejected, and points with confidence below 1.5 or non-finite/degenerate coordinates are discarded before lifting. Object voxels use 0.02 m cells (the same constant used for both the tracker and the visualisation-only global voxel map); the global point cloud $\mathcal{P}$ is voxel-downsampled at 0.05 m with statistical outlier removal.

Canonical projection. The wall band for Manhattan alignment is the $[0.25, 0.75]$ height-percentile slab; the BEV canvas is 1024×1024 at 0.05 m/px, subsequently cropped to Ω_{obs}, whose extent varies by scene (Table 6 reflects this — no two floors occupy the same crop size). Denoising uses a 3×3 morphological opening and drops connected components under 50 px.

Progressive closure (Sec. 3.2). ρ=1 px/iteration, κ=5 px, σ=5 px, T=600, relocation cardinal test c_m=4, relocation cap μ=400 (Dijkstra steps), outlier cutoff k=2.0 applied to clusters of $n_{\min} \geq 4$ seeds. These five $(\rho, \kappa, \sigma, \mu, k)$ are shared by all three seeding variants below and were not re-tuned per variant.

Seeding (Sec. 3.2.2, ablated in Sec. 5.3). FPS draws up to n_s=100 seeds from an eligible pool capped at 20 000 cells with placement cardinal test c_p=4, first point fixed by RNG seed 42 for reproducibility. Camera-pose seeding takes every recorded frame pose on the floor (height tolerance 0.5 m, no subsampling — stride 1) as a candidate seed directly, with no pool or cardinal test, since a frame position is free space by construction. Uniform (random) sampling of the same eligible pool as FPS is implemented (`place_candidates` in `room_detection.py`) but retained only as commented-out code; it produced the closure-sequence panels of Fig. 2b and was not run under the current evaluation harness.

Delineation and sightlines (Sec. 3.2.4–3.3). K=50 rays/seed, seed sightline cap ℓ_s=150 px; delineated regions under 40 px are dropped, and rooms holding fewer than 2000 annotated 3D points are dropped at the point-labelling stage; the interior fill is confined to the footprint eroded by 6 px.

Object-to-room assignment. The two variants compared in Sec. 5.3 (*object surface majority* and *object-seed visibility voting*) were produced by a scoring harness that is not part of this code snapshot — only its outputs (`object_eval/`) are available. We therefore report its results in Sec. 5.1–5.3 without restating its internal thresholds; *object-seed visibility voting* is, by its stored description, the ray-agreement rule of Eq. (6)–(7).

T Systems that Recover a Room Layer

Table 5 groups the systems that maintain a room layer by the input each requires.

Table 5: Systems that recover a room layer, by the input they require. ✓ = operates without that input. Only the last two rows are free of poses, depth and calibration; of those, ours is the only one whose room layer is recovered geometrically, without a learned room-transition prior. *Mono-Hydra is pose-free in the camera sense but requires IMU.

System	RGB only	Pose-free	Calib-free	Room layer from
Kimera [22]	✗	✗	✗	free-space graph
Hydra [14]	✗	✗	✗	places graph, cut at constrictions
HOV-SG [28]	✗	✗	✗	clearance threshold + flooding
Clio [18]	✗	✗	✗	task-driven clustering
OccuSG [9]	✗	✗	✗	occupancy decomposition
Point2Graph [31]		*point cloud input*		border + region detection
Mono-Hydra [27]	✓	✗*	✗	free-space graph
LEXI-SG [15]	✓	✓	✓	learned room-transition prior
Ours	✓	✓	✓	progressive boundary closure

U Published Numbers in Context

Why published numbers are not a comparison. The closest prior systems report room quality on overlapping HM3D scenes, and Table 3 places those figures beside ours for context. They are not

Table 6: Room mIoU by floor, on the top-down grid (Sec. 4.2). n_{pred} is each method's predicted room count (HOV-SG / ours) against that floor's own observed region count. The final row is the mean over floors. Bold: better mIoU per floor. Ours is ahead on 7 of 10; the paired Wilcoxon signed-rank test over these differences gives $p = 0.065$, so the margin is reported as a trend.

Scene	Fl.	n_{gt}	n_{pred}		room mIoU↑	
			HOV-SG	Ours	HOV-SG	Ours
00824-Dd4bFSTQ8gi	0	10	8	11	0.816	**0.887**
00829-QaLdnwvtxbs	0	7	5	7	**0.842**	0.795
00843-DYehNKdT76V	0	7	3	5	0.611	**0.734**
00843-DYehNKdT76V	1	7	5	9	0.692	**0.804**
00873-bxsVRursffK	0	4	2	4	0.350	**0.647**
00873-bxsVRursffK	1	9	6	9	0.798	**0.885**
00877-4ok3usBNeis	0	7	3	8	0.698	**0.859**
00877-4ok3usBNeis	1	9	3	8	**0.755**	0.689
00890-6s7QHgap2fW	0	6	4	8	**0.842**	0.781
00890-6s7QHgap2fW	1	6	5	5	0.936	**0.939**
mean over floors					0.734	**0.802**

a controlled comparison and we do not treat them as one: Hydra and HOV-SG consume posed RGB-D, LEXI-SG's stronger variant consumes ground-truth poses and simulator depth, the published scores are computed over 3D free-space voxels or floor-plane points rather than our grid cells, and they are averaged per scene where we average per floor. Our controlled comparison is the one in Table 2, where every method consumes the identical top-down map produced by our own reconstruction and is scored by the identical harness. LEXI-SG is the nearest work by input assumption — monocular RGB without poses or calibration — and we could not run it directly: its code was public at the time of submission.

V Scene-Wise Analysis

Table 6 and Figure 6 break Table 2's room-mIoU row down by floor. The margin over HOV-SG is not uniform: it is largest on `00873` floor 0 (0.647 vs. 0.350 — HOV-SG collapses an 11-region floor into 2 predicted rooms there, Sec. 5.6) and reversed on `00829` floor 0, `00877` floor 1 and `00890` floor 0, where HOV-SG's more conservative room count happens to land closer to a good IoU match on those particular layouts. Ours is ahead on 7 of 10 floors.

W Ablation Figures

Figure 7 accompanies the ablation of Sec. 5.3.

X Further Experimental Material

X.1 Closure Scale

Per-room closure-iteration is not persisted by the current pipeline, so no closure-scale distribution is reported here; the qualitative closure sequence of Figure 2b illustrates the effect.

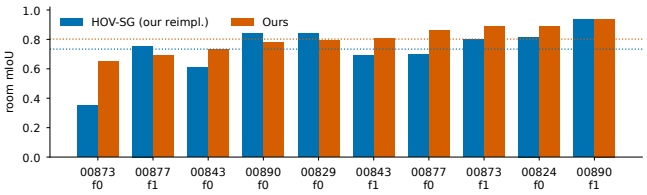


Figure 6: Table 6 plotted per floor, sorted by our mIoU. Dotted lines mark the two methods' mean-over-floors totals from Table 2 (0.734 / 0.802). Per-floor mIoU varies considerably for both methods (HOV-SG: 0.350–0.936; ours: 0.647–0.939), so neither total is representative of every floor.

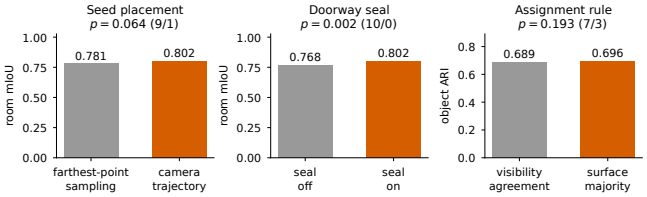


Figure 7: The three ablations of Table 4, each scored on the metric it affects and each with the variant used in the final method in orange: seed placement and the doorway seal on room mIoU, the assignment rule on object ARI. Paired Wilcoxon p and the win/loss split over the ten floors are printed above each panel.

X.2 Runtime

Measured on an AMD Ryzen 7 5800H (8 cores, 16 threads) with 7 GB of RAM, no GPU, single-threaded, over the ten evaluated floors at 0.05 m per cell. Reconstruction is excluded, since on HM3D it is replaced by the rendered walk (Sec. 4.1) and is shared by both methods.

Room detection takes 6.6 ± 3.4 s per floor against 0.3 ± 0.1 s for the HOV-SG baseline on the same map, so our method is roughly twenty times slower. The cost follows from the mechanism: the closure loop performs one dilation and one Euclidean distance transform over the $N \times N$ grid per iteration, giving $O(TN^2)$ for an iteration cap T, where HOV-SG runs a single watershed. The remaining stages are minor by comparison: rasterisation 0.2 s, and the nearest-labelled-point completion applied to both methods 0.8 s. Runtime scales with the reconstruction rather than with the room count, ranging from 3.2 s on the smallest floor (0.80 M points) to 13.4 s on the largest (1.85 M).

Extracting door frames adds 5.5 ± 1.9 s per floor, but this figure is dominated by loading and sampling the annotated mesh once per floor rather than by the hulling itself. In the monocular pipeline the frames arrive as a segmentation output already computed during reconstruction, so the equivalent cost is the projection and convex hull alone.